AQ9.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.
Functions of several variables:

Level 1:

1 point
Which of the following are correct with respect to $f(x, y) = x^2 - 2y$f(x, y) = x2 − 2y, where x, yx, y are real?

- [ ] Domain of ff is R^2R2
- [ ] Domain of ff is RR
- [ ] Range of ff is RR
- [ ] Range of ff is R^2R2

No, the answer is incorrect.
Score: 0
Accepted Answers:
Domain of ff is R^2R2
Range of ff is RR

1 point
Let $D \subset R^n$ D ⊂ Rn and $f: D \longrightarrow R^m$ f : D ⟶ Rm, $g: D \longrightarrow R^m$ g : D ⟶ Rm be multivariable functions on DD. Which of the following statements is(are) TRUE?

- [ ] The product function fgfg is defined on DD by $fg(x\sim) = f(x\sim) \times g(x\sim), x\sim \in D$fg(∼x) = f(∼x) × g(∼x), ∼x ∈ D
- [ ] The sum function $f + g$f + g is defined on DD by $(f + g)(x\sim) = f(x\sim) + g(x\sim), x\sim \in D$(f + g)(∼x) = f(∼x) + g(∼x), ∼x ∈ D
- [ ] Let $c \in R$c ∈ R. The function cfcf is defined on DD by $(cf)(x\sim) = \frac{1}{c} \times f(x\sim), x\sim \in D$(cf)(∼x) = c1 × f(∼x), ∼x ∈ D
- [ ] If m = 1m = 1 and $g(x\sim) \neq 0, x\sim \in D$g(∼x) = 0, ∼x ∈ D, then the function f/gf/g is defined on DD by $(f/g)(x\sim) = f(x\sim)/g(x\sim), x\sim \in D$(f/g)(∼x) = f(∼x)/g(∼x), ∼x ∈ D

No, the answer is incorrect.
Score: 0
Accepted Answers:
The sum function $f + g$f + g is defined on DD by $(f + g)(x\sim) = f(x\sim) + g(x\sim), x\sim \in D$(f + g)(∼x) = f(∼x) + g(∼x), ∼x ∈ D

If m = 1m = 1 and $g(x\sim) \neq 0, x\sim \in D$g(∼x) = 0, ∼x ∈ D, then the function f/gf/g is defined on DD by $(f/g)(x\sim) = f(x\sim)/g(x\sim), x\sim \in D$(f/g)(∼x) = f(∼x)/g(∼x), ∼x ∈ D

1 point

Suppose $f: D \longrightarrow R$f : D ⟶ R is a function defined on a domain $D \subset R$D ⊂ R. Which of the following can be ff?

- [ ] $f(x) = 4 \pm \sqrt{x - 10}$f(x) = 4 ± x − 10
- [ ] $f(x) = \frac{x^2}{x+1}$f(x) = x+1x2
- [ ] $f(x) = log(\sqrt{x^4 + 3})$f(x) = log(x4 + 3)
- [ ] $f(x) = x^3 + x^2 - x$f(x) = x3 + x2 − x

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f(x) = \frac{x^2}{x+1}$f(x) = x+1x2

$f(x) = log(\sqrt{x^4 + 3})$f(x) = log(x4 + 3)

$f(x) = x^3 + x^2 - x$f(x) = x3 + x2 − x

1 point

Which of the following statements is(are) TRUE?

- [ ] The output of a scalar valued multivariable function is always a real number.
- [ ] The output of a vector valued multivariable function can be a real number.
- [ ] The output of a single variable vector valued function is always a vector.
- [ ] The output of a vector valued multivariable function can not be a real number.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The output of a scalar valued multivariable function is always a real number.
The output of a single variable vector valued function is always a vector.
The output of a vector valued multivariable function can not be a real number.

1 point

Consider the function $S(t) = (\cos t, \sin t, 2t)$S(t) = (cos t, sin t, 2t), where tt is real. Choose the set of correct options.

- [ ] SS is a single-variable vector-valued function
- [ ] SS is a multi-variable vector-valued function
- [ ] The domain of SS is RR and the co-domain of SS is R^3R3
- [ ] The domain of SS is R^3R3 and the co-domain of SS is R^3R3

No, the answer is incorrect.
Score: 0
Accepted Answers:
SS is a single-variable vector-valued function
The domain of SS is RR and the co-domain of SS is R^3 R3
How many of the following functions is(are) a vector valued multivariable function?

- $f(x, y) = \frac{1}{3\pi}e^{x^2 + y^2}$ · f(x, y) = 3π1ex2+y2
- $f(x, y) = (x^3, y^2)$· f(x, y) = (x3, y2)
- $f(x, y) = (x + y, x - y, 2xy)$· f(x, y) = (x + y, x − y, 2xy)
- $f(x, y, z) = x^2 + xyz - z^2$ · f(x, y, z) = x2 + xyz − z2
- $f(x) = (x + 1, x - 1)$· f(x) = (x + 1, x − 1)
- $f(x, y, z) = (e^{x + y + z}, 0, 0)$· f(x, y, z) = (ex+y+z, 0, 0)

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point
1 point
Suppose $g: D \longrightarrow R^3$ g : D ⟶ R3 is a function defined on a domain $D \subset R^2$ D ⊂ R2. Which of the following functions may be gg?

☐ $g(x, y) = 4x + 3y - 12xy$g(x, y) = 4x + 3y − 12xy

☐ $g(x, y) = (x^4 + 1, y^4 + 1, xy)$g(x, y) = (x4 + 1, y4 + 1, xy)

☐ $g(x, y, z) = (e^x - e^z, e^z - e^y)$g(x, y, z) = (ex − ez, ez − ey)

☐ $g(x, y) = (sin(x + y), cos(x - y))$g(x, y) = (sin(x + y), cos(x − y))

☐ $g(x, y) = (0, sin(e^x), cos(e^x))$g(x, y) = (0, sin(ex), cos(ex))

No, the answer is incorrect.
Score: 0
Accepted Answers:
$g(x, y) = (x^4 + 1, y^4 + 1, xy)$g(x, y) = (x4 + 1, y4 + 1, xy)
$g(x, y) = (0, sin(e^x), cos(e^x))$g(x, y) = (0, sin(ex), cos(ex))

Level 2:

1 point
Which of the following are correct with respect to $f(x, y) = \sqrt{1 - \frac{x^2}{9} - \frac{y^2}{16}}$f(x, y) = $\sqrt{1 - 9x2 - 16y2}$, where x, yx, y are real?

☐ Domain of ff is $D = \{(x, y) | \frac{x^2}{9} + \frac{y^2}{16} \leq 1\}$D = {(x, y)|9x2 + 16y2 ≤ 1}

☐

Domain of ff is $D = \{(x, y) | \frac{x^2}{9} + \frac{y^2}{16} \geq 1\}$D = {(x, y)|9x2 + 16y2 ≥ 1}

☐

Range of ff is $[0, 1]$[0, 1]

☐

Range of ff is $[0, 1)$[0, 1)

☐

Range of ff is $[1, \infty)$[1, ∞)

No, the answer is incorrect.

Score: 0

Accepted Answers:

Domain of ff is $D = \{(x, y) | \frac{x^2}{9} + \frac{y^2}{16} \leq 1\}$D = {(x, y)|9x2 + 16y2 ≤ 1}

Range of ff is $[0, 1]$[0, 1]

1 point

A function $F: R^2 \rightarrow R^3$F : R2 → R3 has the form $F(x, y) = (P(x, y), Q(x, y), R(x, y))$F (x, y) = (P (x, y), Q(x, y), R(x, y)). Choose the correct option(s).

☐

FF is a multi-variable vector-valued function

☐

FF is a single-variable vector-valued function

☐

P, Q, RP , Q, R are single-variable vector-valued functions

☐

P, Q, RP , Q, R are multi-variable vector-valued functions

☐

P, Q, RP , Q, R are multi-variable scalar-valued functions

No, the answer is incorrect.

Score: 0

Accepted Answers:

FF is a multi-variable vector-valued function

P, Q, RP , Q, R are multi-variable scalar-valued functions

1 point

Consider $f(x, y) = \sqrt{1 - \frac{x^2}{9} - \frac{y^2}{4}}$f(x, y) = 1 − 9x2 − 4y2

$\sqrt{}$

. Note: An ellipse is of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$a2x2 + b2y2 = 1 where a, ba, b are the dimensions of the ellipse.

☐

Ellipses of increasing size represent the curves $\{(x, y) | f(x, y) = c\}${(x, y)|f(x, y) = c}, where cc increases from 00 towards 11

☐

Ellipses of decreasing size represent the curves $\{(x, y) | f(x, y) = c\}${(x, y)|f(x, y) = c}, where cc increases from 00 towards 11

☐

Ellipses of constant size represent the curves $\{(x, y) | f(x, y) = c\}${(x, y)|f(x, y) = c}, where cc increases from 00 towards 11

☐

The curves $\{(x, y) | f(x, y) = c\}${(x, y)|f(x, y) = c}, where $c \in [0, 1)$c ∈ [0, 1), cannot be represented by an ellipse

☐

$f(x, y)$ = 1f(x, y) = 1 is a point

No, the answer is incorrect.

Score: 0

Accepted Answers:

Ellipses of decreasing size represent the curves $\{(x, y) | f(x, y) = c\}${(x, y)|f(x, y) = c}, where cc increases from 00 towards 11

$f(x, y)$ = 1f(x, y) = 1 is a point

1 point

Consider a point source S = (1, 2, 3)S = (1, 2, 3) radiating energy. The intensity I at a given point P = (x, y, z)P = (x, y, z) in space is inversely related to the square of the distance dd between SS and PP : $I(x, y, z) = \frac{k}{d^2}$I(x, y, z) = d2k, where kk is a real positive constant. Choose the correct option(s).

☐ Intensity I is constant on $\{(x, y, z) | (x-1)^2 + (y-2)^2 + (z-3)^2 = 1\}${(x, y, z)|(x − 1)2 + (y − 2)2 + (z − 3)2 = 1} }

☐ Intensity I is constant on $\{(x, y, z) | (x-1)^2 + (y-2)^2 + (z-3)^2 = 2\}${(x, y, z)|(x − 1)2 + (y − 2)2 + (z − 3)2 = 2}}

☐ Intensity I is constant on $\{(x, y, z) | x^2 + y^2 + z^2 = 1\}${(x, y, z)|x2 + y2 + z2 = 1}

☐ Intensity I is constant on $\{(x, y, z) | x^2 + y^2 + z^2 = 2\}${(x, y, z)|x2 + y2 + z2 = 2 }

No, the answer is incorrect.

Score: 0

Accepted Answers:

Intensity I is constant on $\{(x, y, z) | (x-1)^2 + (y-2)^2 + (z-3)^2 = 1\}${(x, y, z)|(x − 1)2 + (y − 2)2 + (z − 3)2 = 1} }

Intensity I is constant on $\{(x, y, z) | (x-1)^2 + (y-2)^2 + (z-3)^2 = 2\}${(x, y, z)|(x − 1)2 + (y − 2)2 + (z − 3)2 = 2}}